

Linear Algebra I for Mathematical Sciences
Mock Term Test – Vectors Spaces and Dependence
Winter 2025

First (Given) Name: _____

Last (Family) Name: _____

Signature: _____

Student Number: _____

Important Instructions

• Pages without QR codes will not be graded.

If you need extra space for solutions, then there is blank space on pages 11-12.

- Do NOT begin until you are instructed to do so.
- Show ALL your work and justify your answers for each question.
- The term test is 110 minutes long.
- No aids are allowed.
- Electronic aids of any kind are forbidden.
- The booklet contains a total of 12 pages.
Please check to ensure that no pages are missing from your test booklet.
- Do NOT remove any pages from this test booklet.
- Do NOT write on the QR codes.
- Have your student card ready for inspection.
- Show us what you learned in the course. Enjoy writing this term test.

On this term test, you may assume that $\mathbb{R}$ and $\mathbb{C}$ are fields.

Q1. (10 points) Consider the following vector space.

$$V = (\mathbb{R}^n, \boxplus, \boxdot) \text{ where } x \boxplus y = (x_1 + y_1, \dots, x_n + y_n) \text{ and } \lambda \boxdot x = (\lambda x_1, \dots, \lambda x_n)$$

Define 3 axioms of a vector space and verify that $(\mathbb{R}, \boxplus, \boxdot)$ satisfies these 3 axiom.

Q2. Determine if the following subsets are vector spaces. If so, provide a proof; if not, provide a counterexample.

(a) (5 points) The real 2 by 2 trace zero matrices

$$\left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}(\mathbb{R}) \mid a + d = 0 \right\}$$

(b) (5 points) Real 2 by 2 singular matrices

$$\left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}(\mathbb{R}) \mid ad - bc = 0 \right\}$$

Q3. (10 points) In any vector space V , the equality $\mathbf{x} + \mathbf{y} = \mathbf{x} + \mathbf{z}$ implies $\mathbf{y} = \mathbf{z}$ where $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$.

Q4. (10 points) Consider the vector space

$$P_3(\mathbb{R}) = \{f(x) \in \mathbf{F}(\mathbb{R}) \mid f(x) \text{ is a polynomial of degree at most } 3\}.$$

Determine if $x^3 - 3x + 3 \in \text{span}(\{x^3 + x^2, x^2 + x, x^3 + 1\})$.

Q5. (10 points) Let V be a real vector space and $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$. Show that $\{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ is linearly independent if and only if $\{\mathbf{x} + \mathbf{y}, \mathbf{y} + \mathbf{z}, \mathbf{z} + \mathbf{x}\}$ is linearly independent.

Q6. (10 points) Let W be the set of real 2 by 2 symmetric matrices. Show that W is a subspace. Then, create a finite set S such that $W = \text{span}(S)$.

Q7. Consider the following system of equations.

$$\begin{cases} x + y + z = 4 \\ 2x + 3y + 2z = 7 \\ 3x + 5y + 3z = 10 \end{cases}$$

(a) (2 points) Construct the appropriate augmented matrix to represent this system of equations.

(b) (5 points) Use elementary row operations to solve for the echelon form of this matrix. List the free and basic variables.

- (c) (3 points) Solve for the solution set for this system of equations.

Q8. (10 points) Suppose that W_1 and W_2 are subspaces of V such that $W_1 \cap W_2 = \{\mathbf{0}\}$. Prove: If $S_1 \subset W_1$ and $S_2 \subset W_2$ are linearly independent, then $S_1 \cup S_2$ is linearly independent.

This page is for additional graded work.

If any work on this page is to be graded, indicate this **CLEARLY** on the question page.

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This is the end of the term test.